Project Design Phase-I
Solution Architecture

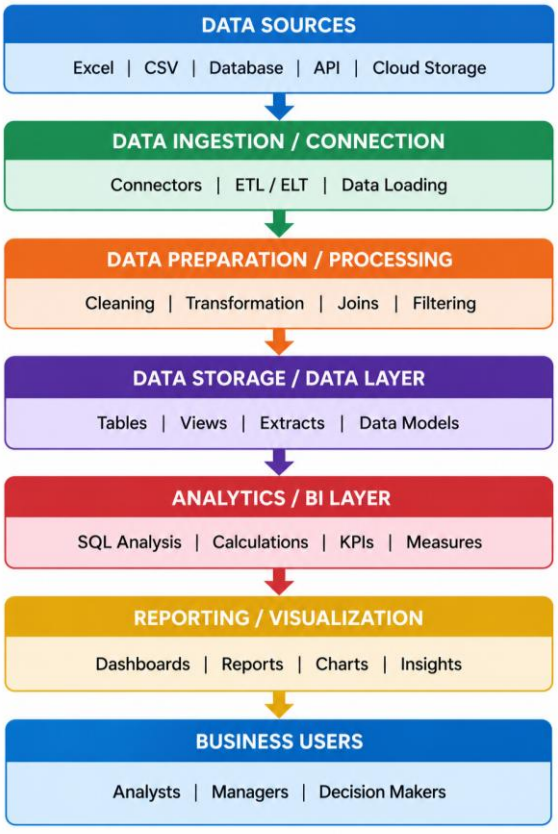
Date	19 September 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Health360: Executive Analytics & Population Health Dashboard
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Sample Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
1. Data Sources	Stores raw, structured healthcare dataset containing 5,000 anonymized patient records across clinical vitals, lifestyle habits, and demographics.	Flat File CSV (smart_healthcare_dataset .csv)
2. Data Ingestion / Connection	Ingests flat CSV data into the analytical environment, validating field data types and establishing schema mapping.	Tableau Desktop Native Connector
3. Data Preparation / Processing	Cleanses dataset and generates pre-calculated measures: Age Category classification (Children to Elder), Calculated Risk Score, and chronic disease binary aggregations.	Tableau Calculated Fields Syntax
4. Data Storage / Data Layer	Converts raw rows into a high-speed columnar data extract to optimize query execution and eliminate live CSV processing bottlenecks.	Tableau Data Extract (.hyper engine)
5. Analytics / BI Layer	Computes dynamic summary metrics, calculates baseline KPIs (Total Patients = 5,000, Baseline Risk = 1.57), and evaluates multi-variable risk matrices.	Tableau Desktop Analytics Engine
6. Reporting / Visualization	Renders 9 specialized visual worksheets, a single-screen Master Dashboard with interactive cross-filtering, and a 9-point guided Tableau Story.	Tableau Desktop Worksheets & Dashboard Canvas
7. Business Users / Endpoints	Provides zero-footprint web access for decision-makers (Dr. Michael Carter, Emily Rodriguez, Dr. Priya Nair) to perform interactive cohort exploration and strategic decision-making.	Tableau Public Web Server / Cloud Engine